

Collective Robotics

Part 8: Cyborg Swarms

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Cyborg

short for cybernetic organism,
has organic and biomechatronic body parts

(Cyborgs and Space in Astronautics (September 1960) by Manfred E. Clynes and Nathan S. Kline)

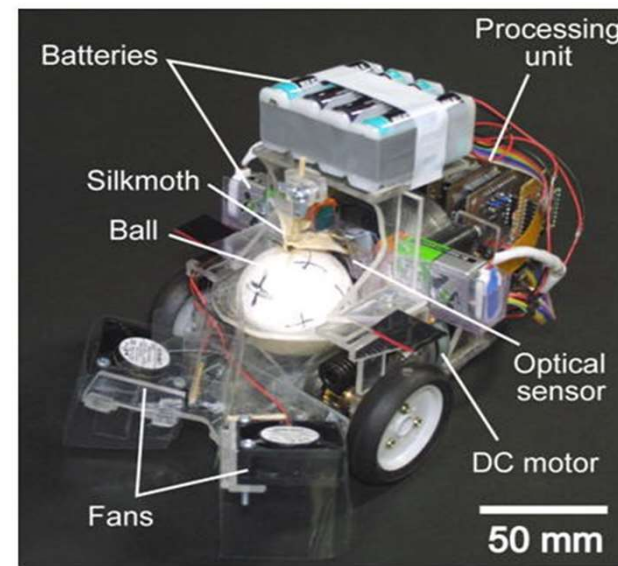
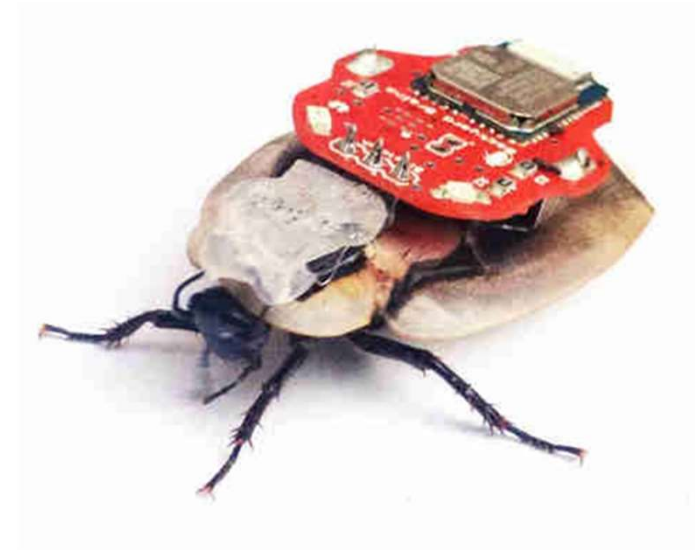
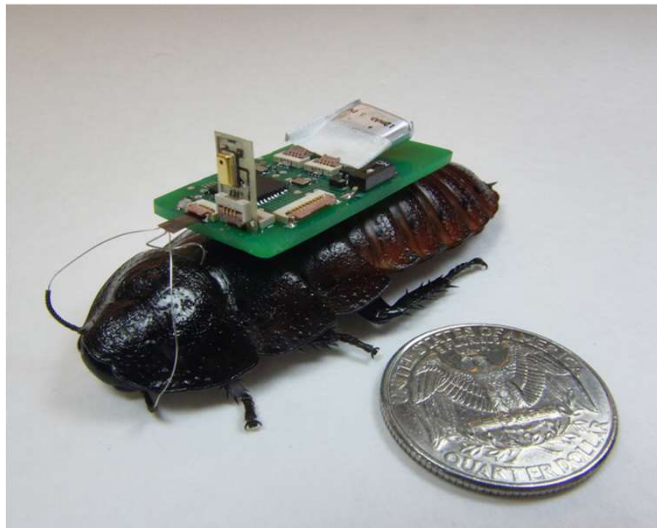
here body parts are swarm members: natural
organisms and robots



Group Discussion

- How can we introduce the concept of cyborg to collective robotics?

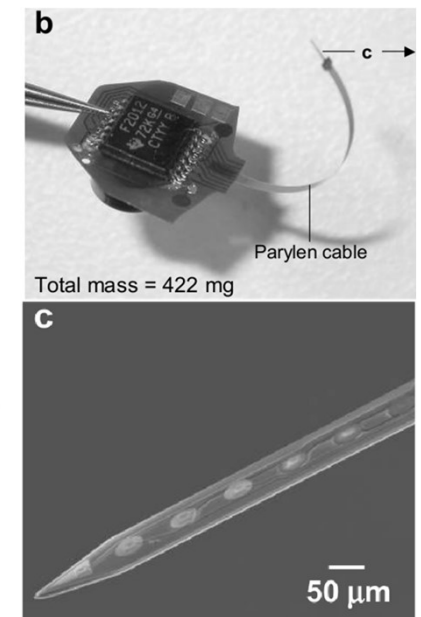
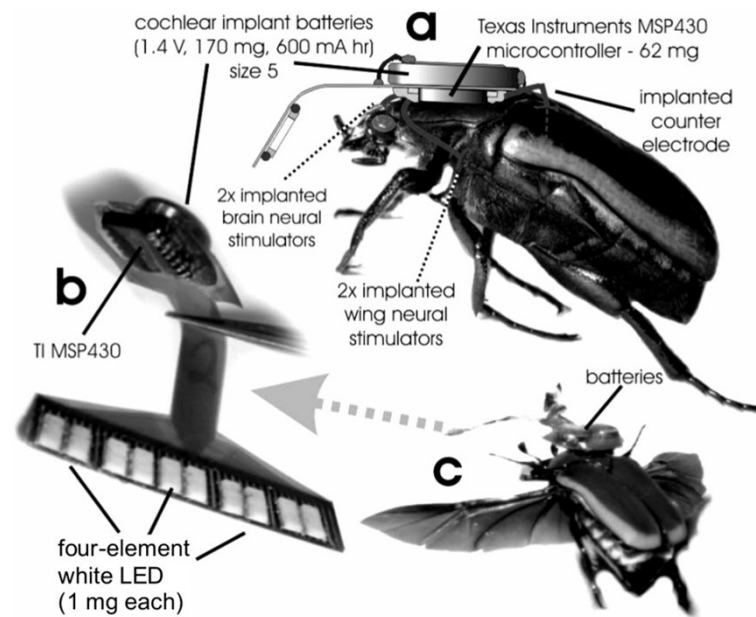
Some real cyborgs



Cyborg beetle (1/3)

Sato et al. (2008)

implantable flight control
microsystem for a cyborg
beetle



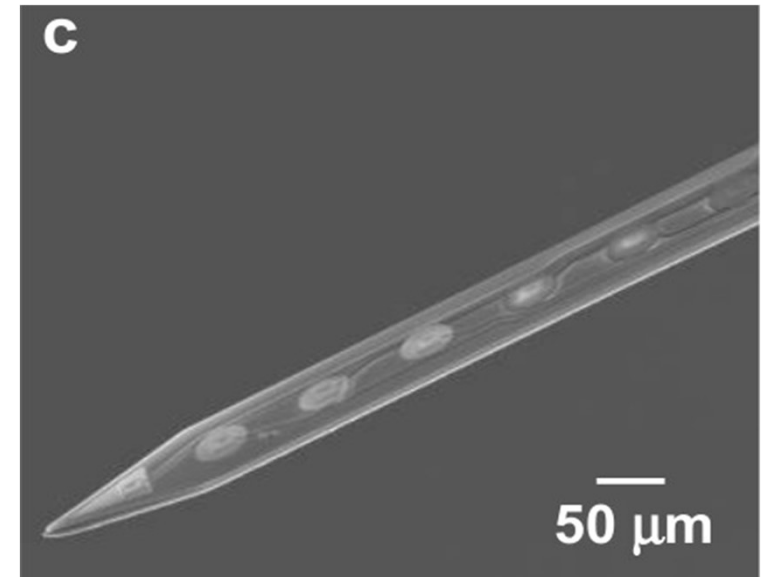
Cyborg beetle (2/3)

neural stimulation system

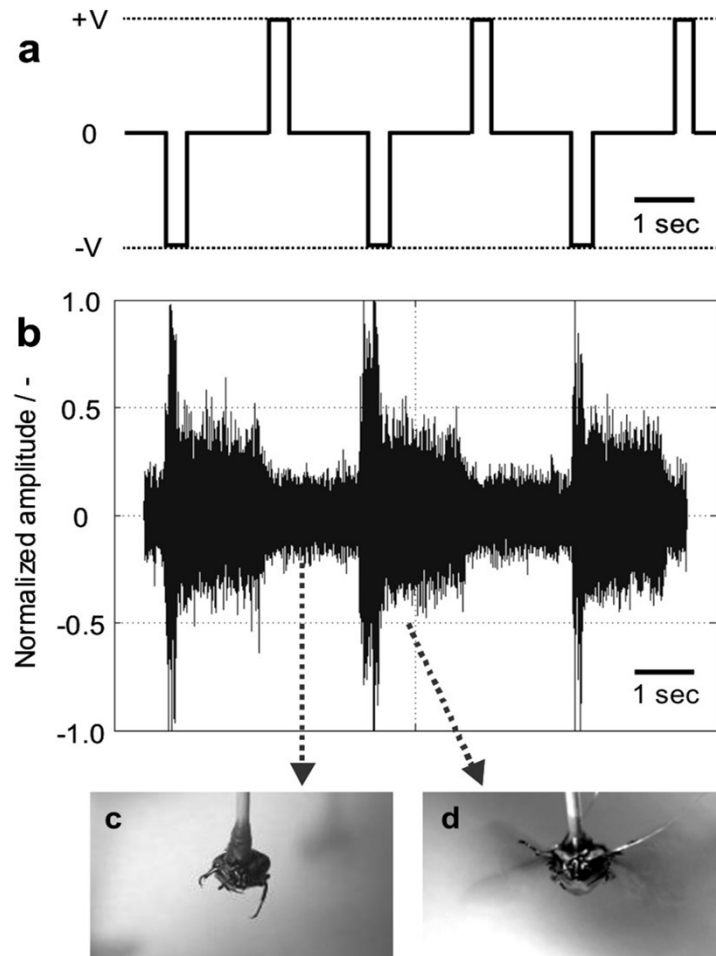
4 neural stimulators (silver thin wire electrodes), implanted:

1 behind eye in flight control area of brain

2 on either side under flight muscles

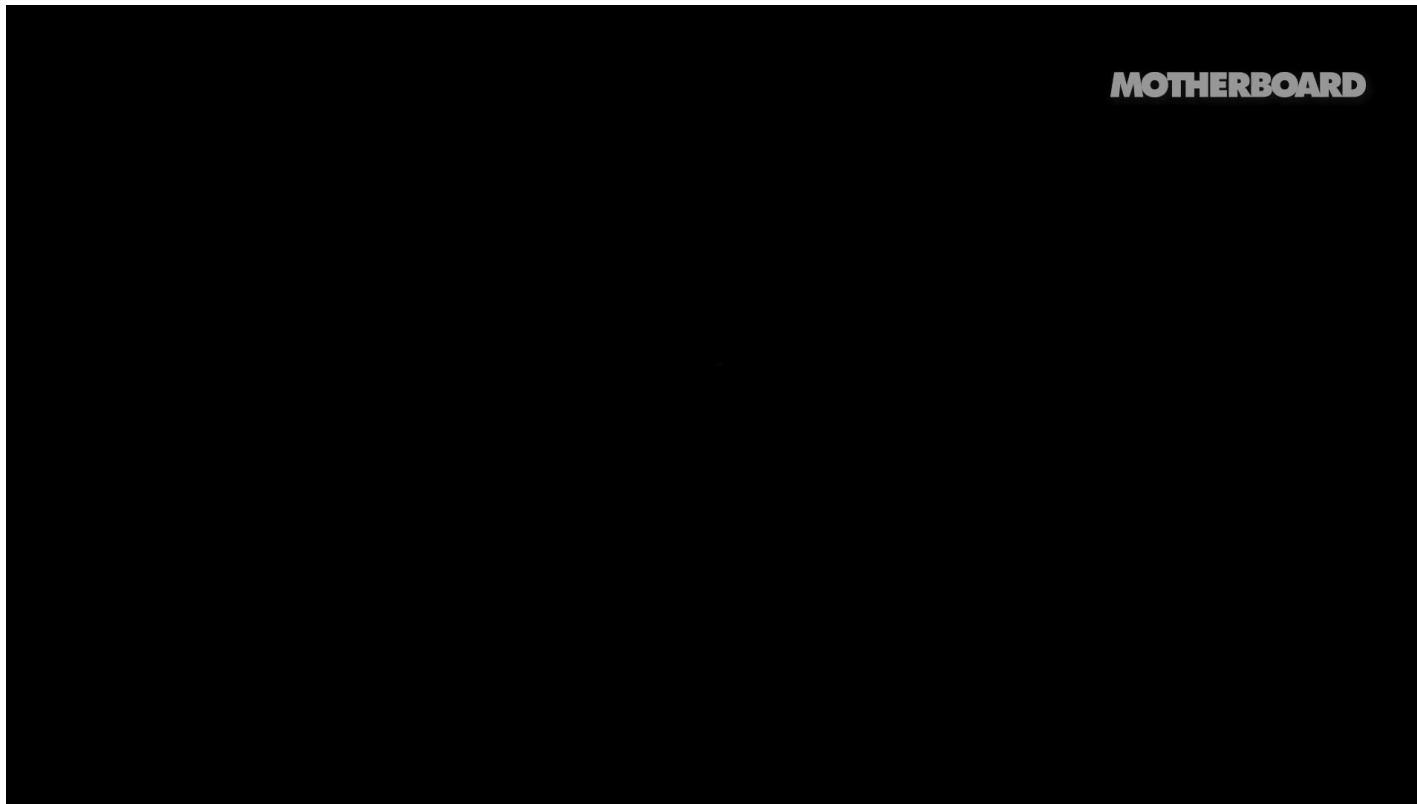


Cyborg beetle (3/3)



- neural stimulation waveform
- wingbeats were recorded on audio tracks
- positive pulses stop flight oscillator,
- negative pulses start oscillator

Cyborg beetle



Project Assisi

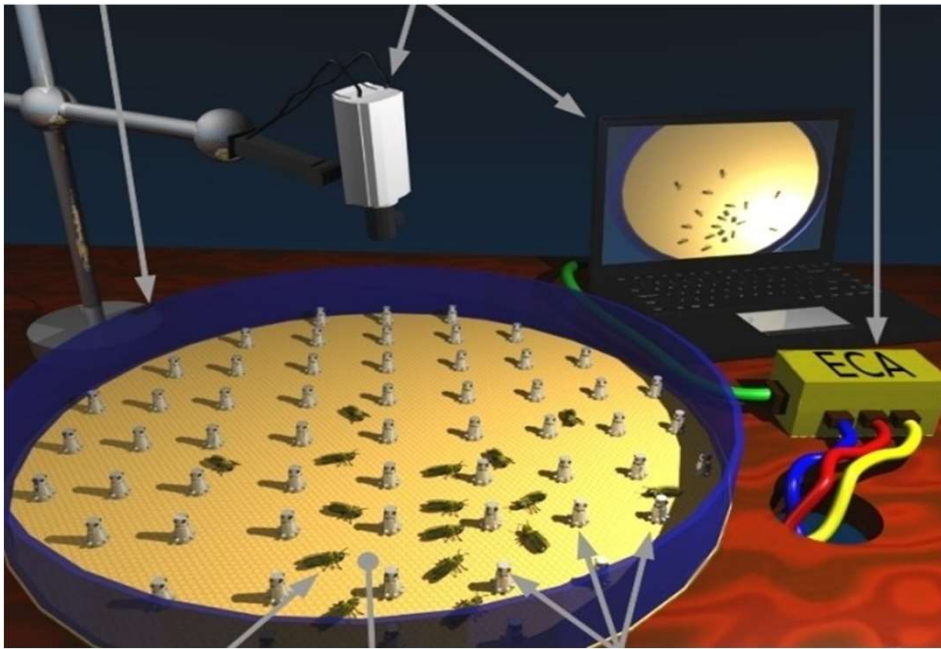
Animal and robot Societies Self-organise and Integrate by Social Interaction (bees and fish)

scientific goals

1. Develop robots that can influence the collective behaviours of animals (bees and fish).
2. Establishing an adaptive and self-organising society build by robots and animals.
3. Enable the robots to autonomously 'learn' the social language of the animals.
4. Establish mixed societies that pursue a common goal which can be defined by human 'users' of the system.
5. Allow the robots to gain novel skills by incorporating the capabilities of the animals (sensors, cognition).



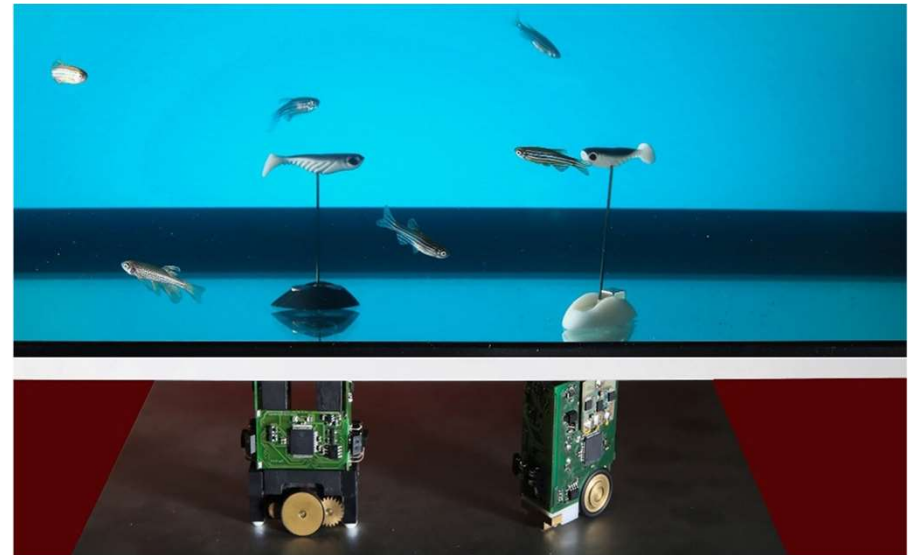
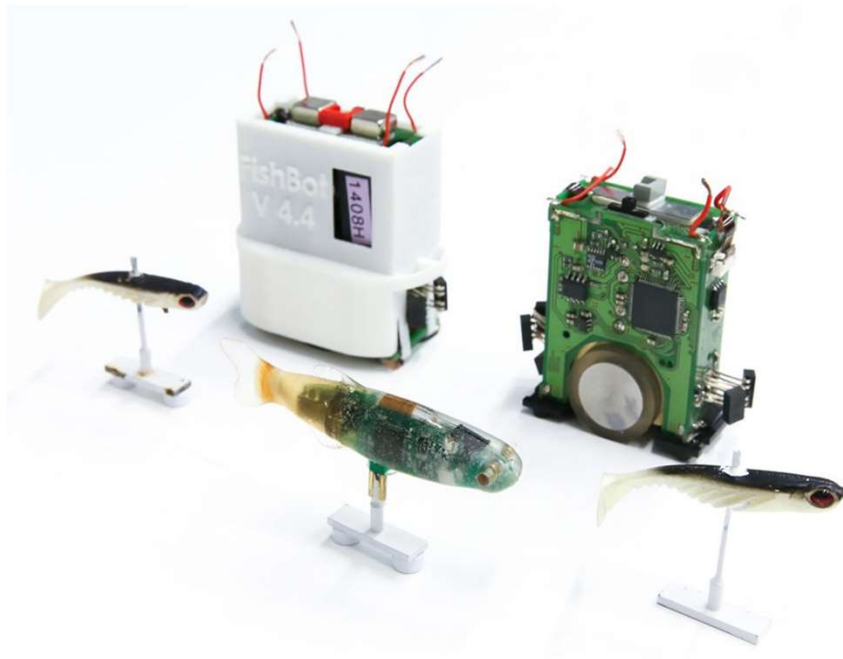
Assisi- honeybees



attraction with temperature, vibrations



Assisi- fish



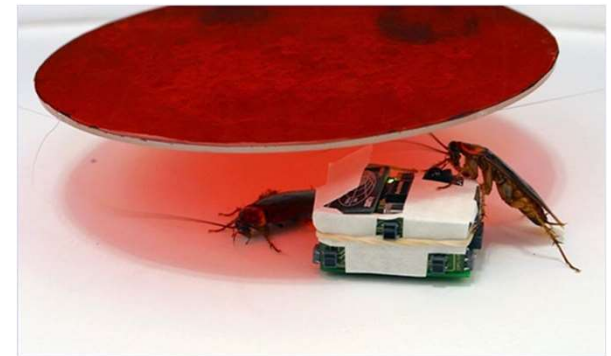
Assisi : Video



Cockroaches and robots

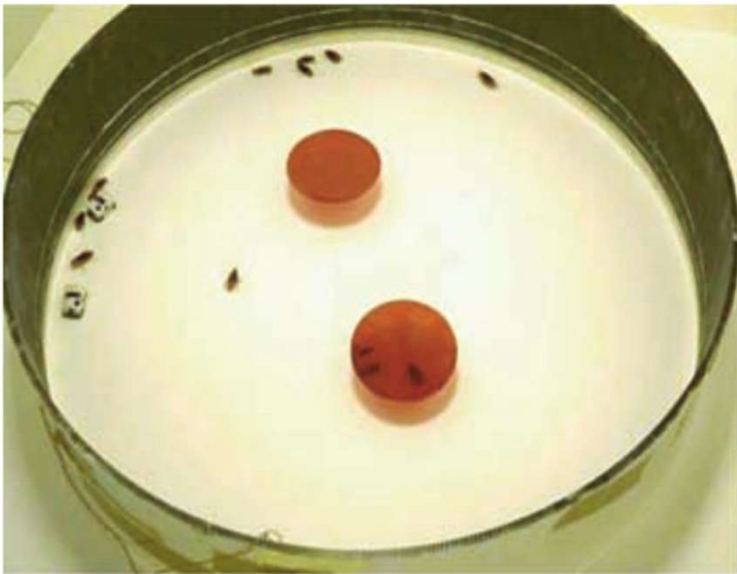
collective decision-making by mixed groups of cockroaches and socially integrated autonomous robots, leading to shared shelter selection

even when in the minority, robots can modulate the collective decision-making process and produce a global pattern not observed in their absence (Halloy et al., 2007)



<http://science.sciencemag.org/content/sci/318/5853/1155.full.pdf>

Experimental setup



circular arena, 2 shelters

- 2 identical shelters,
each large enough to host the entire group
→ cockroaches choose collectively to rest under one
- One shelter is darker
→ cockroaches select the darker shelter by amplifying their individual preference through interindividual interactions
Self-organized choice, no leadership

Chemical signal

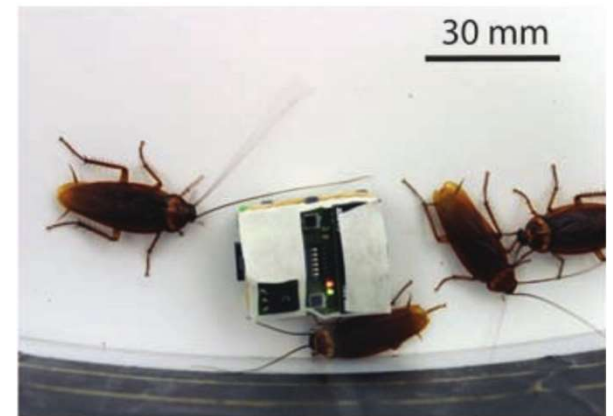
acceptance of robots:

correct chemical signal and behave appropriately

identification of the main molecules constituting the odor that carries cockroach identity

odor collected from male cockroaches

natural and artificial agents equally attractive



Group Discussion

- How to model this behaviors?

Behavioral model – rate equations

$$dx_i/dt = R_i x_e - Q_i x_i \quad i = 1, 2$$

$$dr_i/dt = R_{ri} r_e - Q_{ri} r_i \quad i = 1, 2$$

$$C = x_e + x_1 + x_2$$

$$M = r_e + r_1 + r_2$$

x_i : number of cockroaches in shelter i

r_i : number of robots in shelter i

C : total number of cockroaches

M : total number of robots

x_e, r_e : agents outside the shelters

R, Q : functions for rate per individual of entering or quitting shelters

Behavioral model – rates

cockroach, rate of entering a shelter:

$$R_i = \mu_i (1 - (x_i + \omega r_i)/S_i)$$

robot, rate of entering a shelter:

$$R_{ri} = \mu_{ri} (1 - (x_i + \omega r_i)/S_i)$$

cockroach, leaving a shelter:

$$Q_i = \frac{\theta_i}{1 + \rho \left(\frac{x_i + \beta r_i}{S_i} \right)^n}$$

robot, leaving a shelter:

$$Q_{ri} = \frac{\theta_{ri}}{1 + \rho_r \left(\frac{\gamma x_i + \delta r_i}{S_i} \right)^{n_r}}$$

μ_i, μ_{ri} : max. rate of entering

θ_i : max. rate of leaving

ω : surface of 1 robot as a multiple of the surface of 1 insect

S_i : carrying capacity of shelter i

ρ, n : 'conspecifics'

γ, β, δ : mutual influence

Parameter sets and meaning

Idea of R_i and R_{ri} : crowding effect,
decrease with ratio num.

Individuals in shelter i and its carrying capacity S_i

Number of cockroaches	$Q (s^{-1})$	Number of observations
1	0.066	779
2	0.041	246
3	0.017	137
4	0.008	95
5	0.004	82
6	0.003	69

Table S3. Probability of leaving a light shelter (Q) in relation to the number of cockroaches under the shelter.

When both shelters identical:

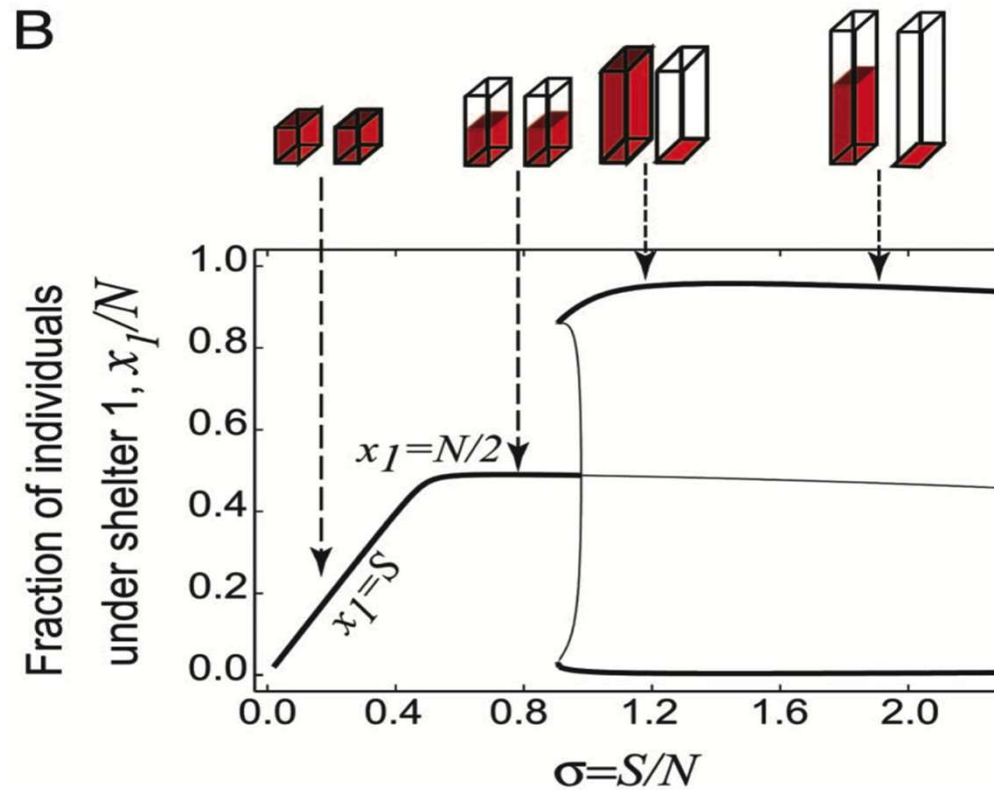
$$S_1 = S_2; \mu_1 = \mu_2; \mu_{r1} = \mu_{r2}; \theta_1 = \theta_2; \theta_{r1} = \theta_{r2}$$

When one shelter is darker than the other: $\theta_1 \neq \theta_2; \theta_{r1} \neq \theta_{r2}$

prob. of entering shelter (μ) independent of dark/light

The higher the pheromone concentration on the robots, the higher the value of β
(influence of robots on cockroaches for quitting rate Q_i)

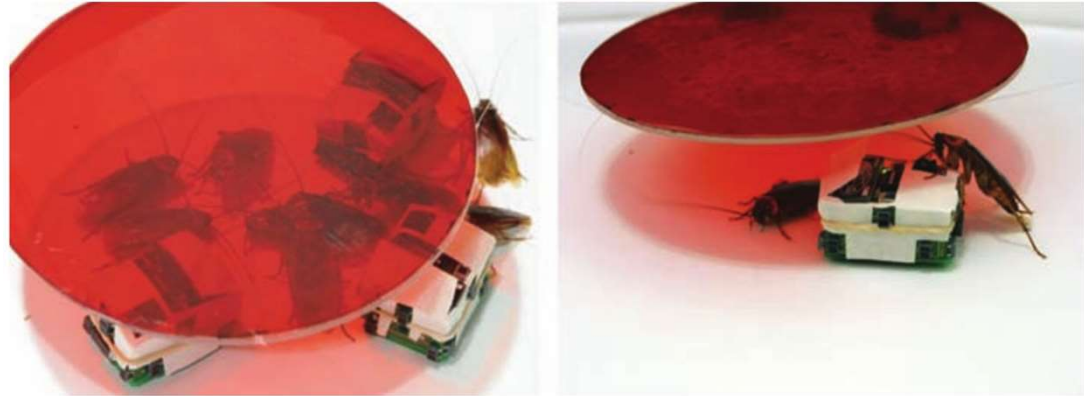
What happens for different shelter capacities?



see [Ame' et al. \(2006\)](#)

$$\frac{dx_i}{dt} = \mu x_e \left(1 - \frac{x_i}{S} \right) - \frac{\theta x_i}{1 + \rho \left(\frac{x_i}{S} \right)^n}$$

Influenced by robots

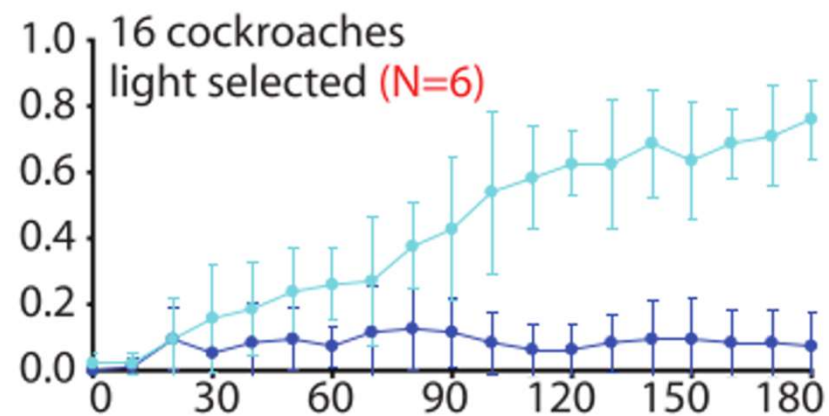
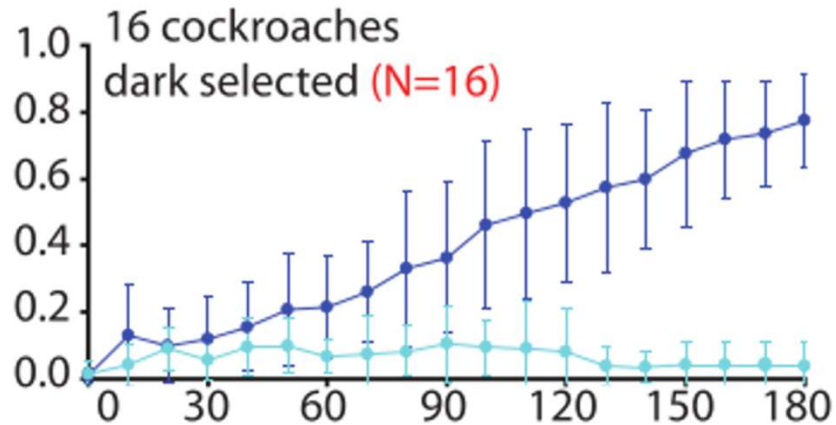


robots programmed to prefer the lighter shelter contrary to cockroaches, same behavioral model but swapping parameters controlling response to darkness,

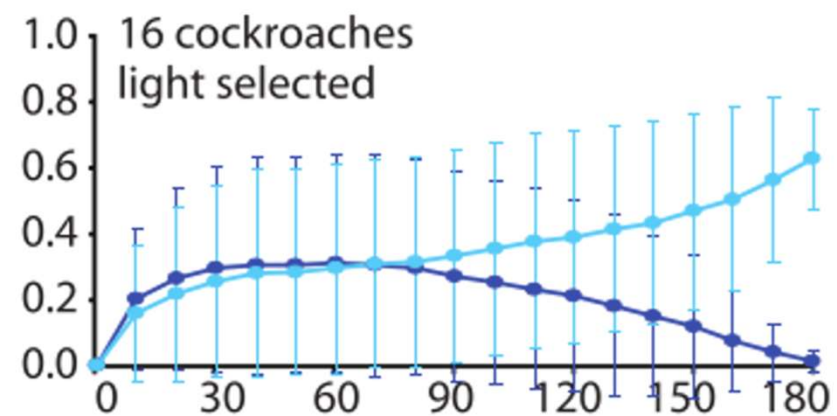
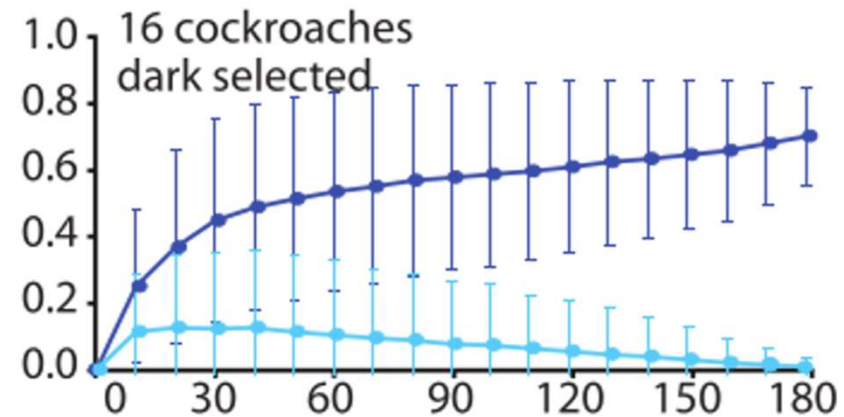
robots induced change of global pattern: inverting the collective shelter preference, lighter shelter chosen 61%

versus only 27% without robots

Selected results



Vertical: Fraction of Population under shelter
Horizontal: time (minutes)



Left: Stochastic Simulations (with noise)
Right: model (rate equation)

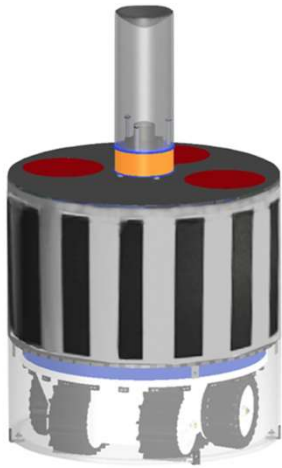
Example of interactions between cockroaches & robots



Collective choice between light/dark shelter



Mixed Societies of Robots and Vertebrates



(a) The design of the Poul-Bot robot. The plexiglas bumper protects chicks fingers from been stuck in the robot's tracks



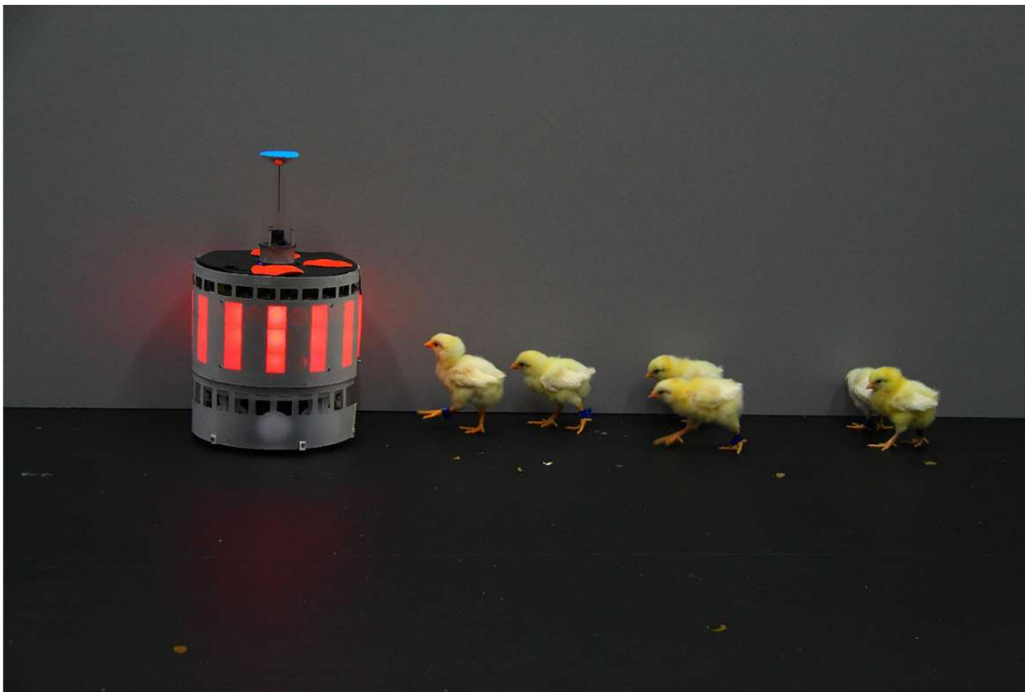
(b) The current version of the robot. Black stripes pattern serves to improve the imprinting efficiency



Chicken Robot project

filial imprinting: 'shortly after hatching we present the moving and vocalizing robot to chicks and they learn that the robot is the mother' ([Gribovskiy et al., 2010](#))

Chicken Robot



Application in the management of farm animals

Color LEDs pattern module and a protective bumper

Array of 16 microphones

Visual & acoustic communication channels - most important for birds

Two minutes of thinking

- 1 minute for highlight
- 1 minute for challenge

References

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